Finite State Machine (FSM) and PLC

Introduction

JAL

Objectives

By the end of the class the student should;

- Describe the operation of a Finite State Machine by identifying states and transitions.
- Be able to represent sequential events as a FSM and program it using ladder logic diagram.

Help Homer to wake up



Draw a ladder logic diagram that makes a red light blink and a buzzer sound intermittently using timers to set a desired frequency. Draw the Finite State Machine and the time diagram.

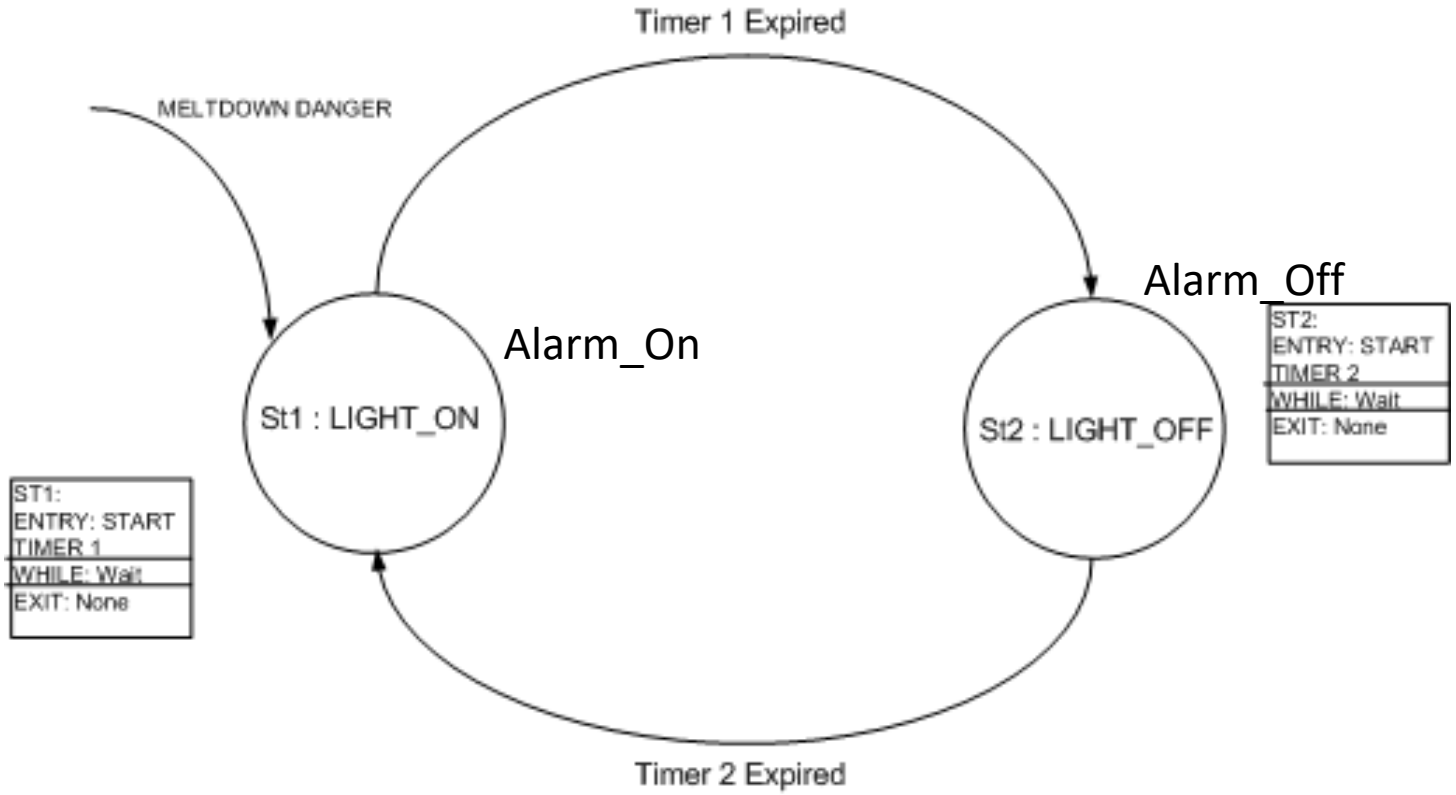
Finite State Machine

- An FSM is a set of states and transitions that model the operating stages of a particular process and its purpose is to automate the process.
- A state is a stage of the process
- A transition is the reason to move from one state to another.
- An FSM is modeled with a graphical representation

Finite State Machine

- There are many graphical representations for FSM. They are intuitive and self-explanatory.
- We just need to be clear and unambiguous with our FSM model.
- Particular software packages have their own FSM representation, all very similar and intuitive. From the model, the software can generate the code to be downloaded in the target.
- As we are considering modern PLCs, we will use commands and techniques recommended to implement FSMs with PLC.

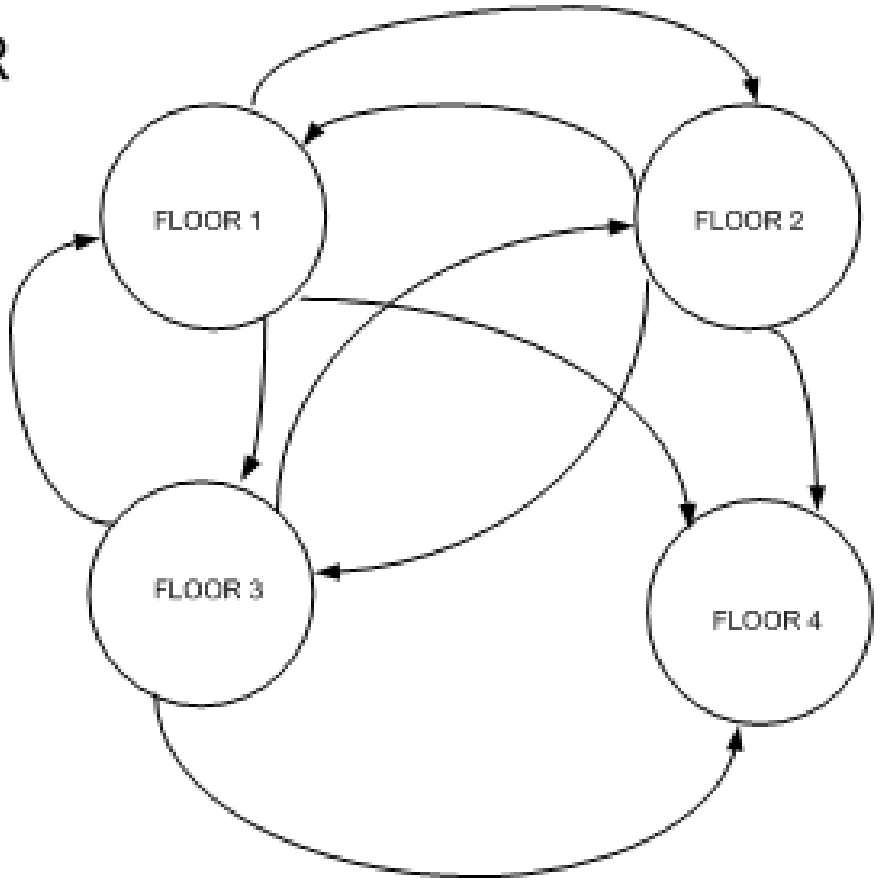
Finite State Machine



ENTRY: Actions that need to be done at entering the state
WHILE: Actions that need to be done during the state
EXIT: Actions that need to be done at exiting the state

Finite State Machine

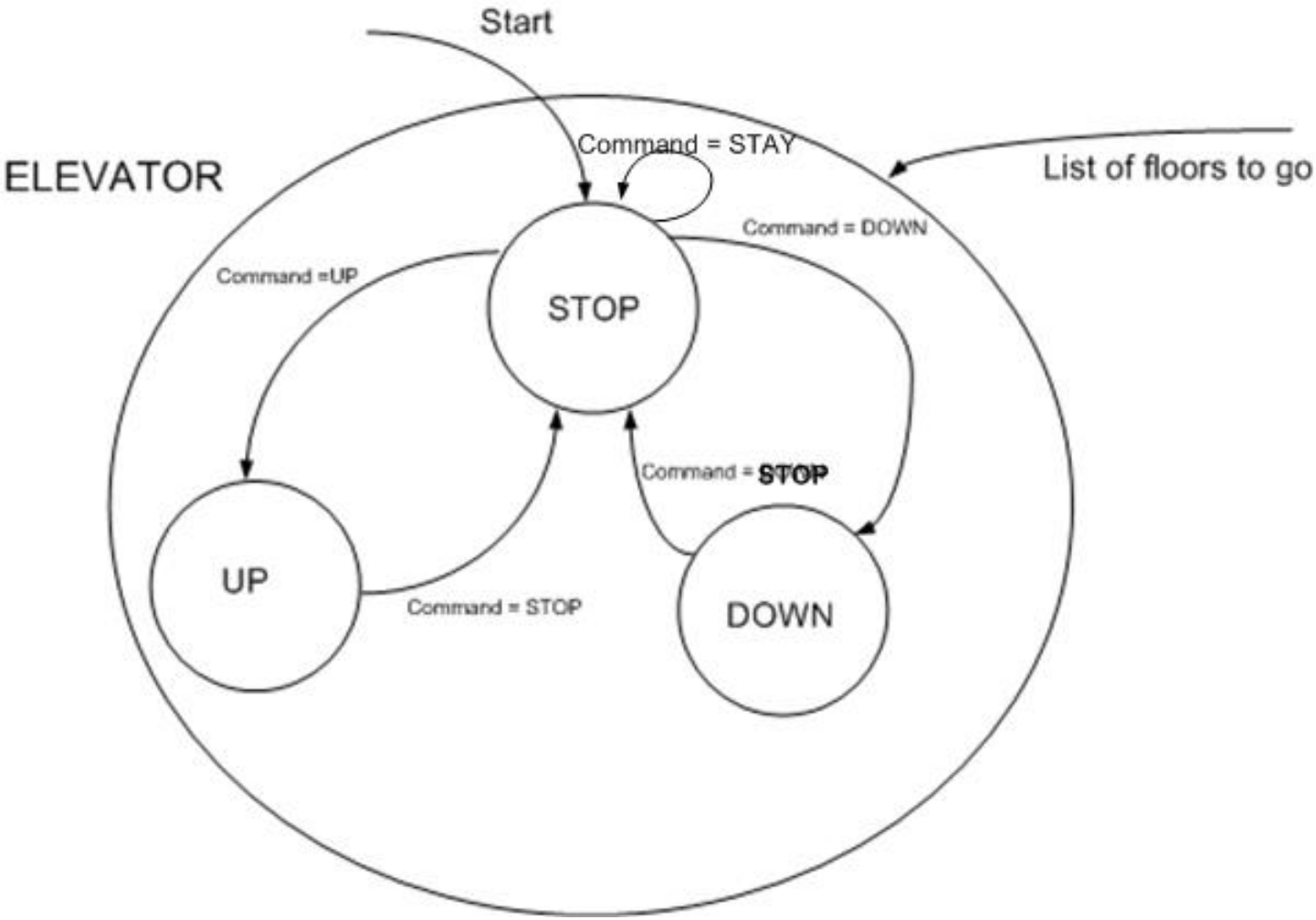
ELEVATOR WITH 4 FLOORS



Finite State Machine

- The previous FSM is not good. It was intended to illustrate the following points:
- The state selection is not unique. The number of states and the definition of the states is up to the designer.
- The selection of states can make a big difference. It can make the design easy or difficult.
- Always strive for intuitive and self-explanatory FSM diagrams. Simplicity is the key. It must be easier to understand and implement. For example, in the case of the elevator…

Finite State Machine



Finite State Machine

STATE: STOP
ENTRY:
1. Brake motor
2. Open door

WHILE:
1. Target = next in list;
2. if Target == empty, then command = STAY
3. if Target.> current floor, then command = UP
4. if Target < current floor, then command = DOWN

EXIT:
1. Close door
2. Command = command

Finite State Machine

STATE: UP
ENTRY:
1. Start up-motion
WHILE:
1. If floor == Target, then command = STOP
2. Else, keep on moving
EXIT:
1. Command = command

Finite State Machine

STATE: DOWN
ENTRY:
1. Start down-motion
WHILE:
1. If floor == Target, then command = STOP
2. Else, keep on moving
EXIT:
1. Command = command

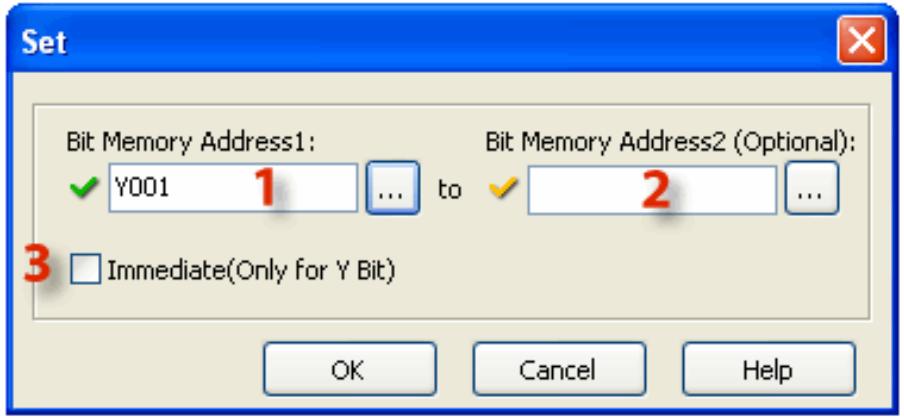
Finite State Machine

- Observation: FSMs designed the way we just saw goes very well when we program a microcontroller or have a SW that allows this kind of sequential representation.
- We will learn a technique to program FSMs using PLCs. There are special commands that allow us to do this. They are the SET and RESET in Click, and Latch and Unlatch in AB which we will use to set or reset a state in the FSM.

The Set Coil instruction

Definition

The **Set** instruction turns **ON** the associated **Bit Memory** when the status of the rung is **True**. The **Bit Memory** stays on after the rung becomes **False**. The **Set** instruction can turn **ON** more than one **Bit Memory** at the same time.



✔ **Entry required or invalid entry** ✔ **Optional** ✔ **Valid entry**

Setup

1 **Bit Memory Address1:** Choose a **Bit Memory Address** to associate with the **Set** instruction. Either type the **Bit Memory Address** directly in the address field or use the **Browse Button** ... to open the **Address Picker.**

2 **Bit Memory Address2 (Optional):** The **Set** instruction permits a consecutive range of bit memory addresses to be **Set** simultaneously. Choose a **Bit Memory Address** to identify the end of the range of **Bit Memory Addresses** or leave this field blank if you only want just one **Bit Memory Address** to respond to this instruction.

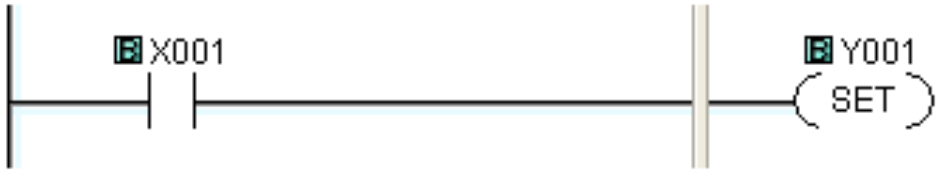
3 **Immediate (Only for Y Bit):** Choose **Immediate** if you want to turn on the **Output(s)** immediately. This feature is only available for actual **Y outputs**. This option is grayed out until a **Y** address is chosen. The **Immediate Icon** will appear adjacent to the **Set Coil** in the **Ladder Editor**.

Note: Each **Set Coil** in your **Ladder Program** should be accompanied by a **Reset Coil** to turn **OFF** the addresses that were turned **ON** by the **Set Coil**.

The Set Coil Instruction

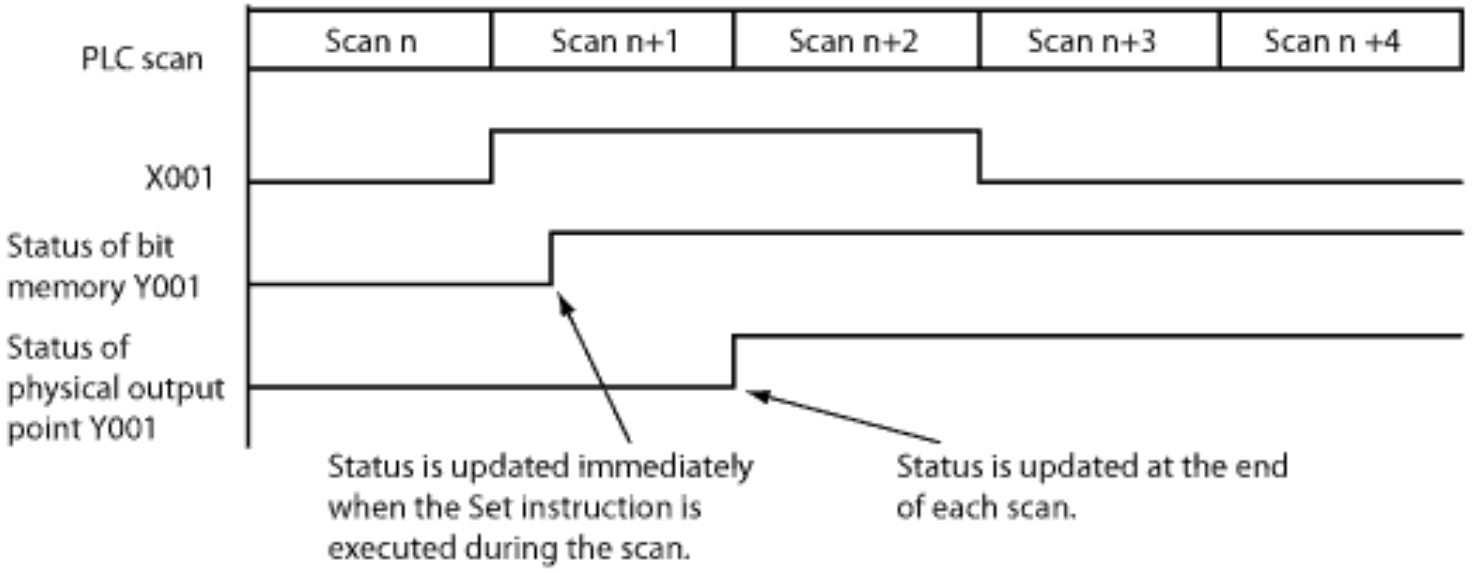
Example Program 1: Default

In the following example, when input status bit **X001** is **ON**, the Set instruction turns **ON** the output status bit **Y001** immediately. After that, **Y001** stays **ON** even if **X001** is turned **OFF**.





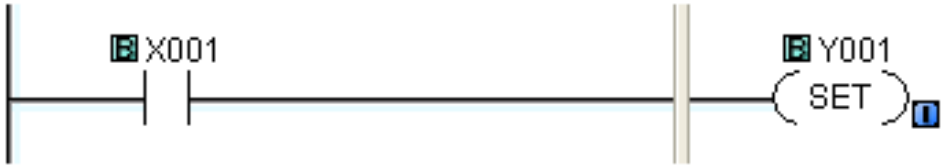
Timing Diagram



The Set Coil instruction

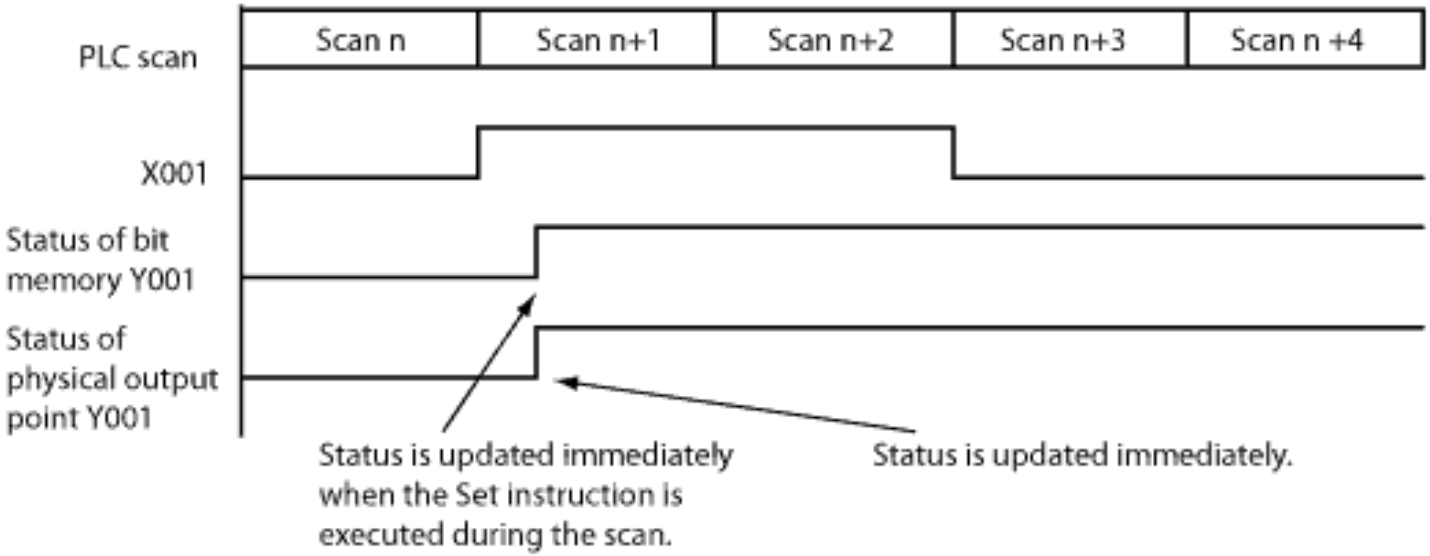
Example Program 2: Immediate (Available only for the Y memory type)

In the following example, when input status bit **X001** is **ON**, the **Set** instruction turns **ON** both the output status bit **Y001** and the physical output point (**Y001** is the 1st output point on the **CPU** module) immediately. Please refer to the timing diagram to see the difference.





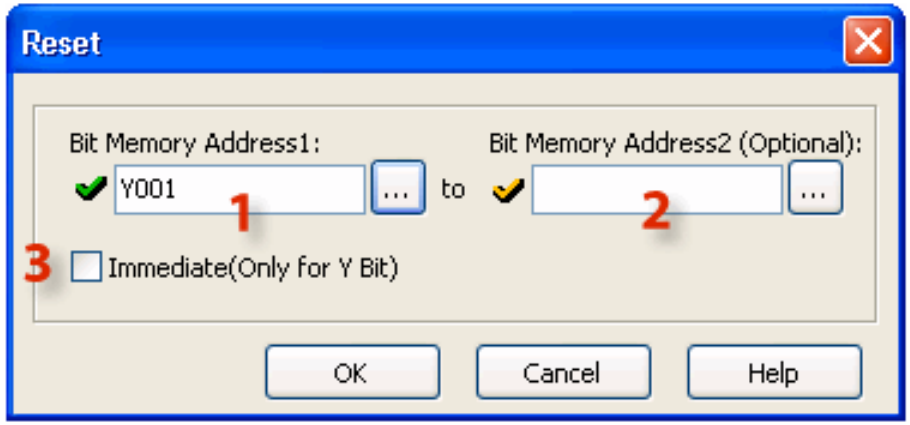
Timing Diagram



The Reset Coil instruction

Definition

The **Reset** instruction shall turn off the associated bit memory when the status of the rung is true. The Bit Memory stays OFF after the rung becomes False. A **Reset** instruction shall turn off more than one bit memory at the same time.



✔ **Entry required or invalid entry** ✔ **Optional** ✔ **Valid entry**

Setup

1 **Bit Memory Address 1:** Choose a **Bit Memory Address** to associate with the **Reset** instruction. Either type the **Bit Memory Address** directly in the address field or use the **Browse Button** ... to open the **Address Picker.**

2 **Bit Memory Address 2:** The **Reset** instruction permits a consecutive range of bit memory addresses to be **Reset** simultaneously. Choose a **Bit Memory Address 2** to identify the end of the range of **Bit Memory Addresses** or leave this field blank if you only want just one output to respond to this instruction.

3 **Immediate (Only for Y Bit):** Choose the **Immediate Icon** if you want to turn **OFF** the **Output(s)** immediately. This feature is only available for actual **Y outputs**. This option is grayed out until a **Y** address is chosen. The **Immediate Icon** will appear adjacent to the **Reset Coil** in the **Ladder Editor**.

Note: Each **Set Coil** in your **Ladder Program** should be accompanied by a **Reset Coil** to turn **OFF** the addresses that were turned **ON** by the **Set Coil**.

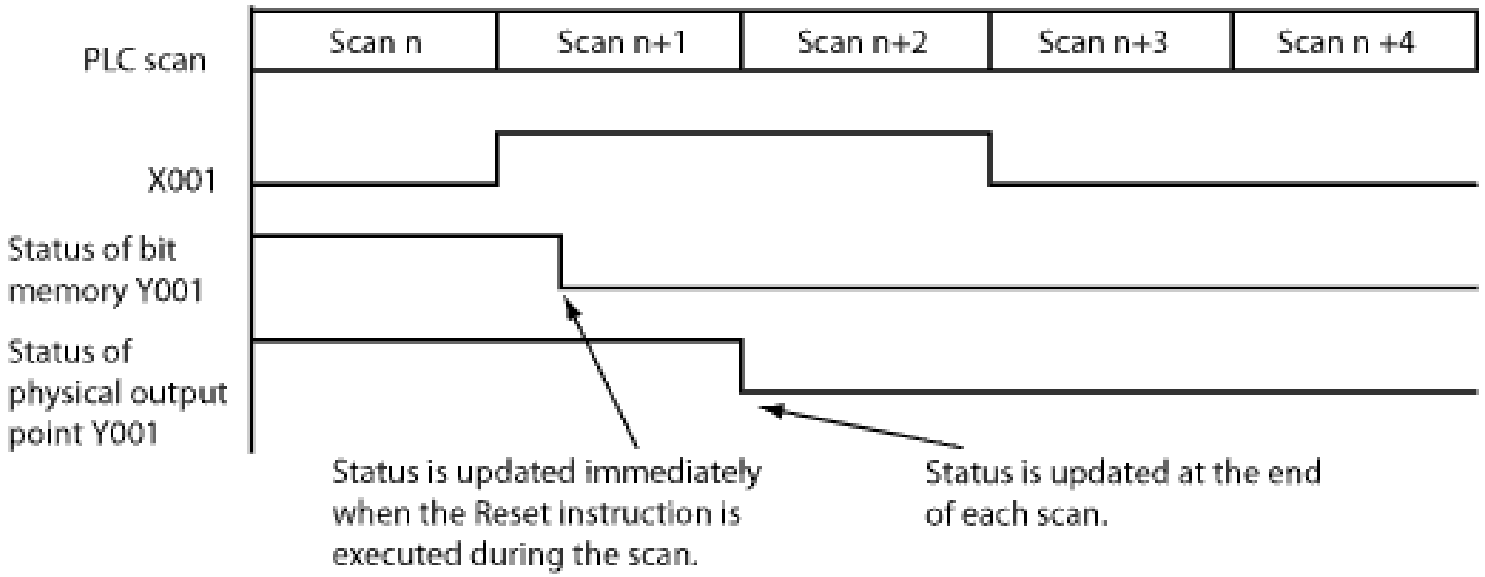
The Reset Coil Command

Example Program 1: Default (No option)

In the following example, when input status bit **X001** is **ON**, the **Reset** instruction turns **OFF** the output status bit **Y001** immediately. **Y001** stays **OFF** even if **X001** is turned **OFF**.



Timing Diagram



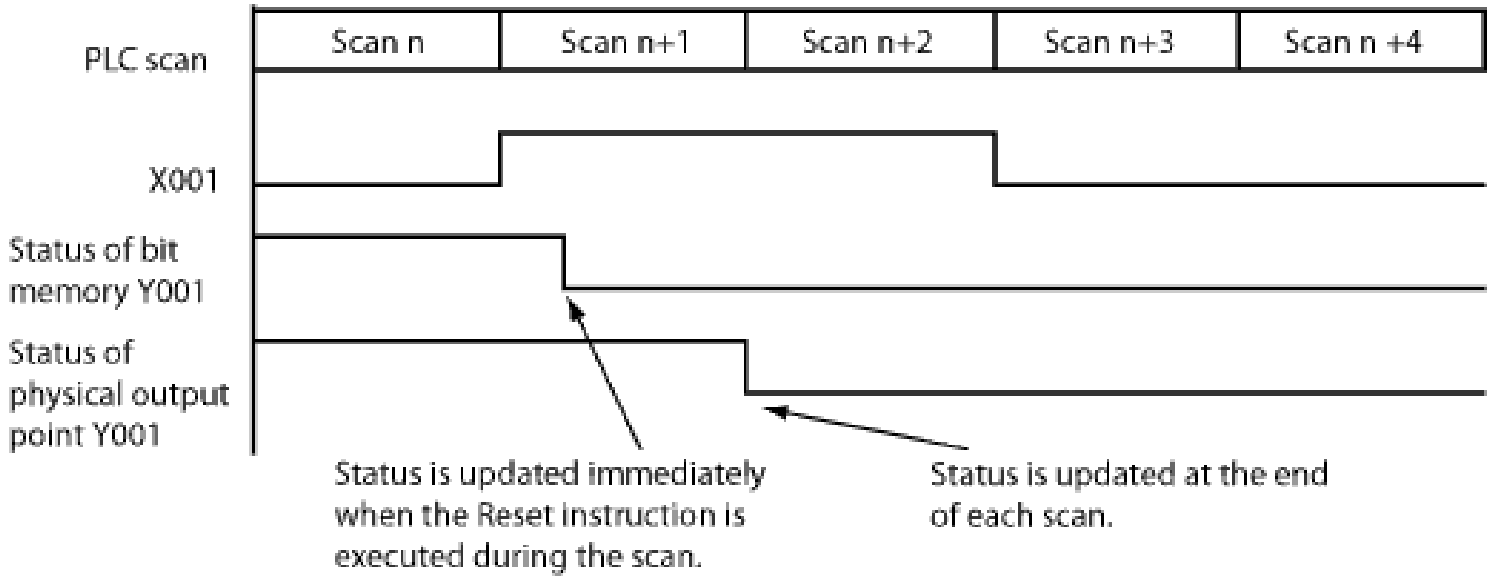
The Reset Coil command

Example Program 1: Default (No option)

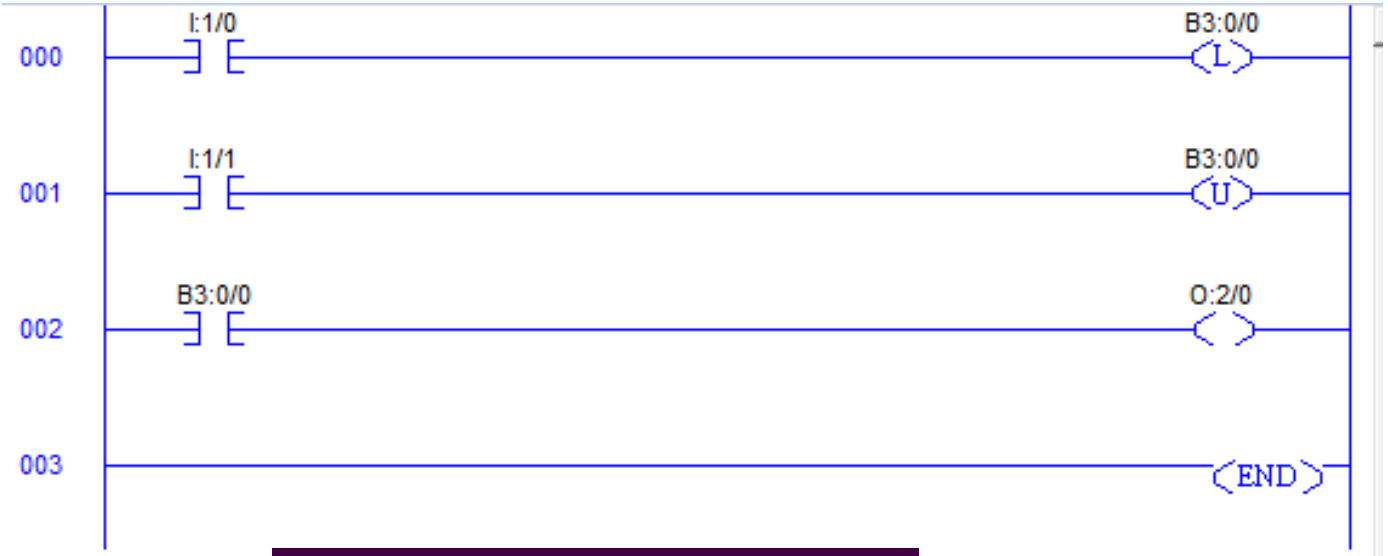
In the following example, when input status bit **X001** is **ON**, the **Reset** instruction turns **OFF** the output status bit **Y001** immediately. **Y001** stays **OFF** even if **X001** is turned **OFF**.

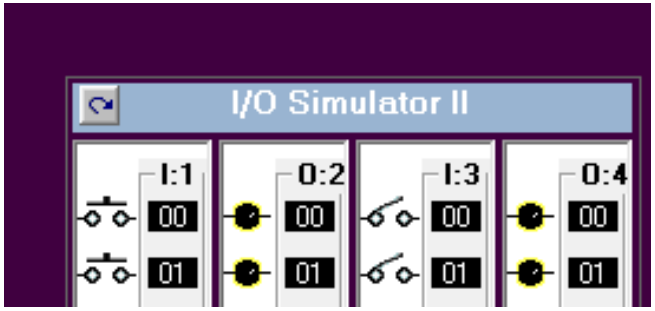


Timing Diagram



Latch/Unlatch Command In AB

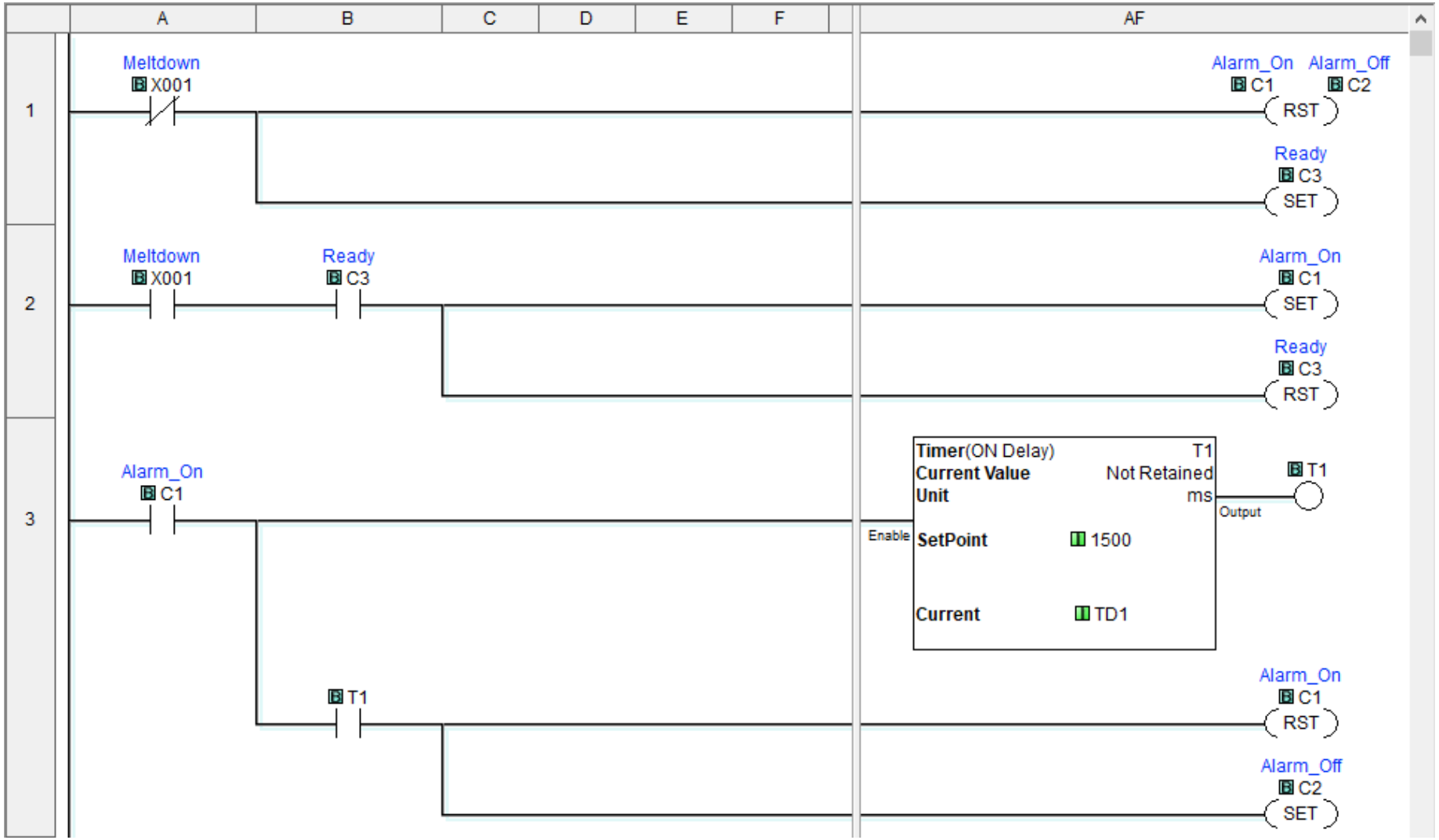




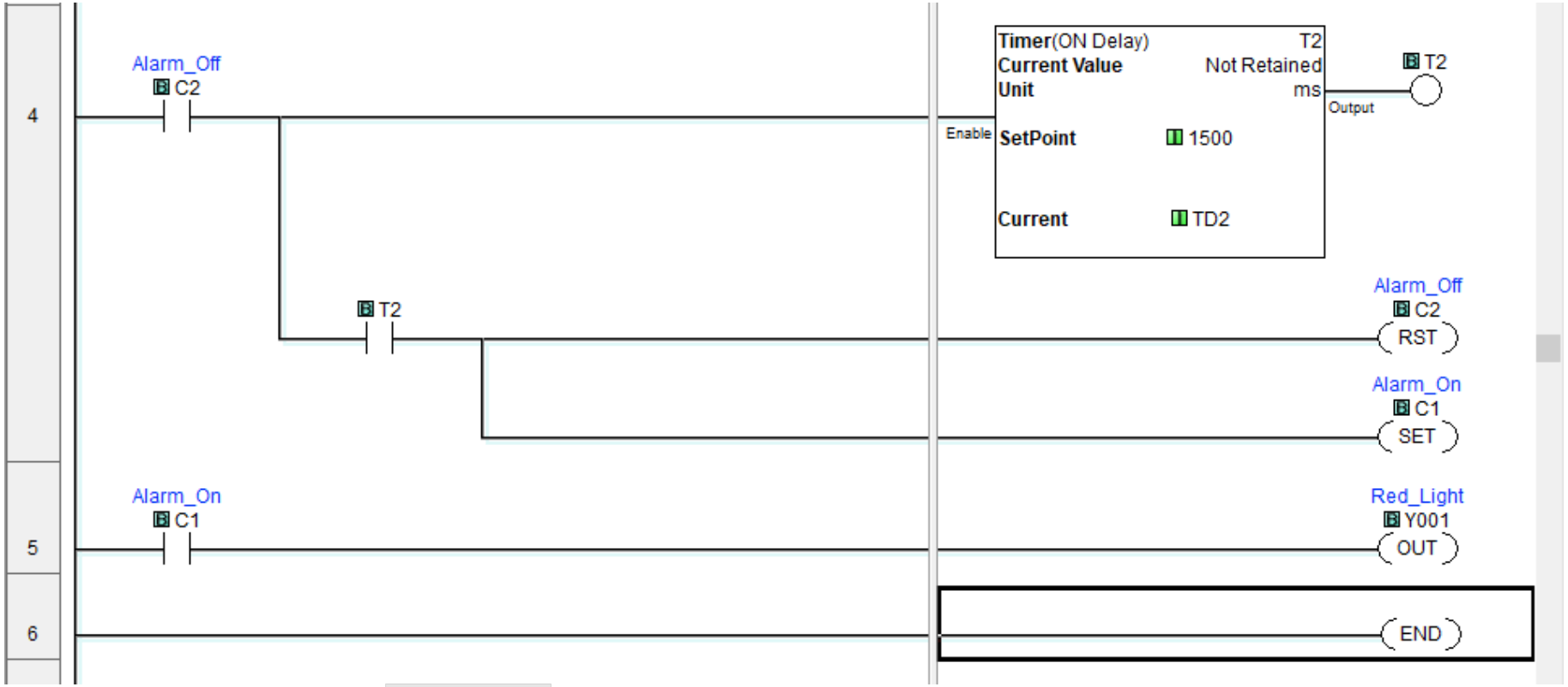
Example 1: Homer's Alarm

- Implement the Meltdown alarm using Click PLC programming. Specifications:
- In normal conditions the red light must be off.
- When the meltdown condition is on, the red light must flash with an interval of 1.5s.
- When the meltdown condition is gone and the conditions are normal again, the FSM must reset itself and get ready again for another eventuality.

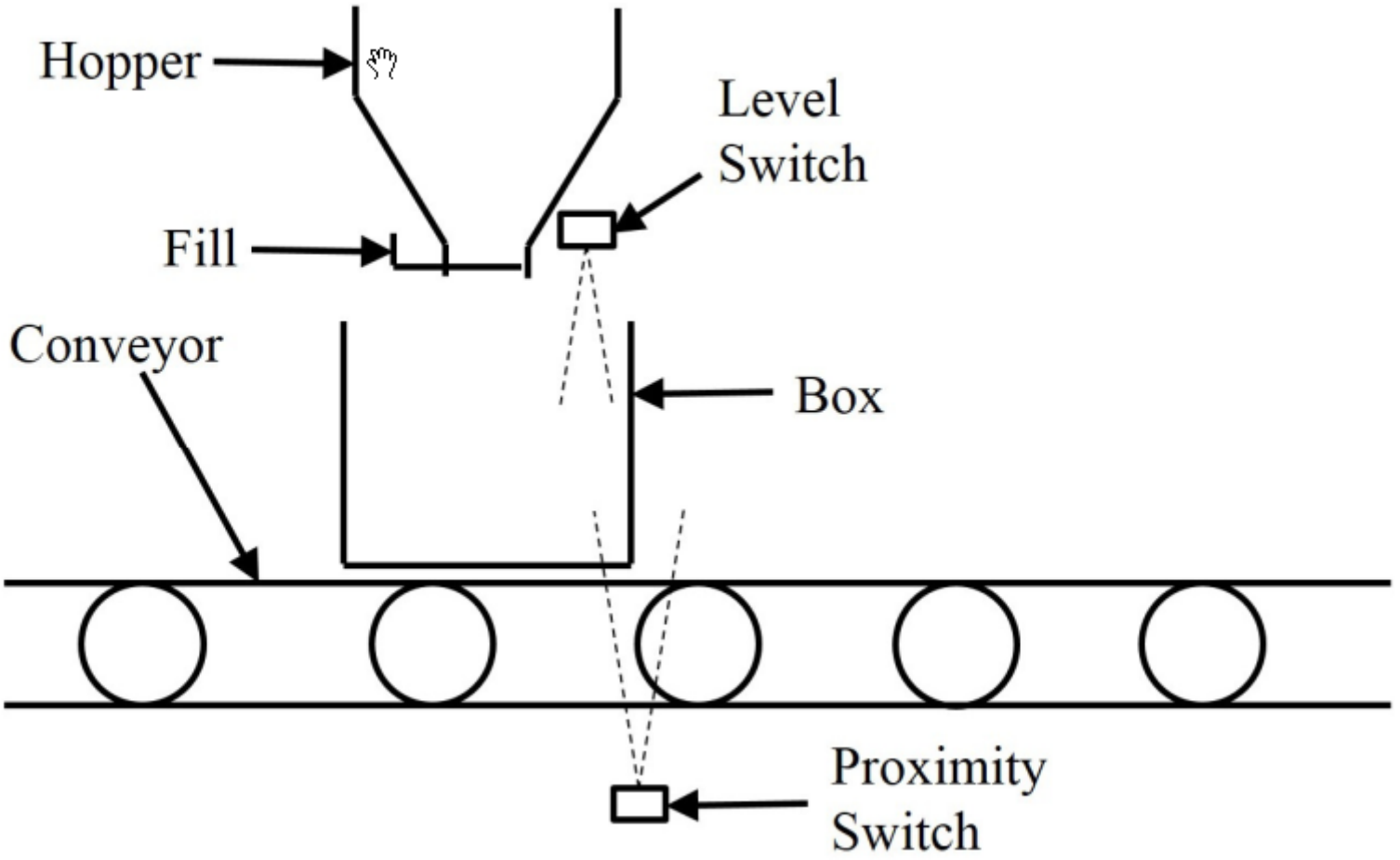
Homer’s alarm solution



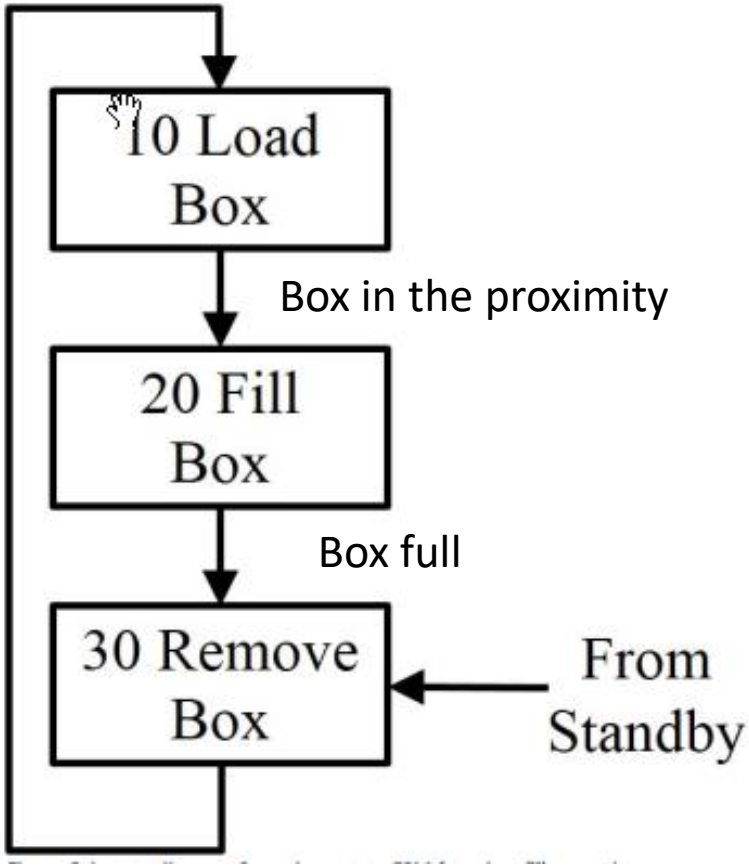
Homer's alarm solution



Example 2: Box filling



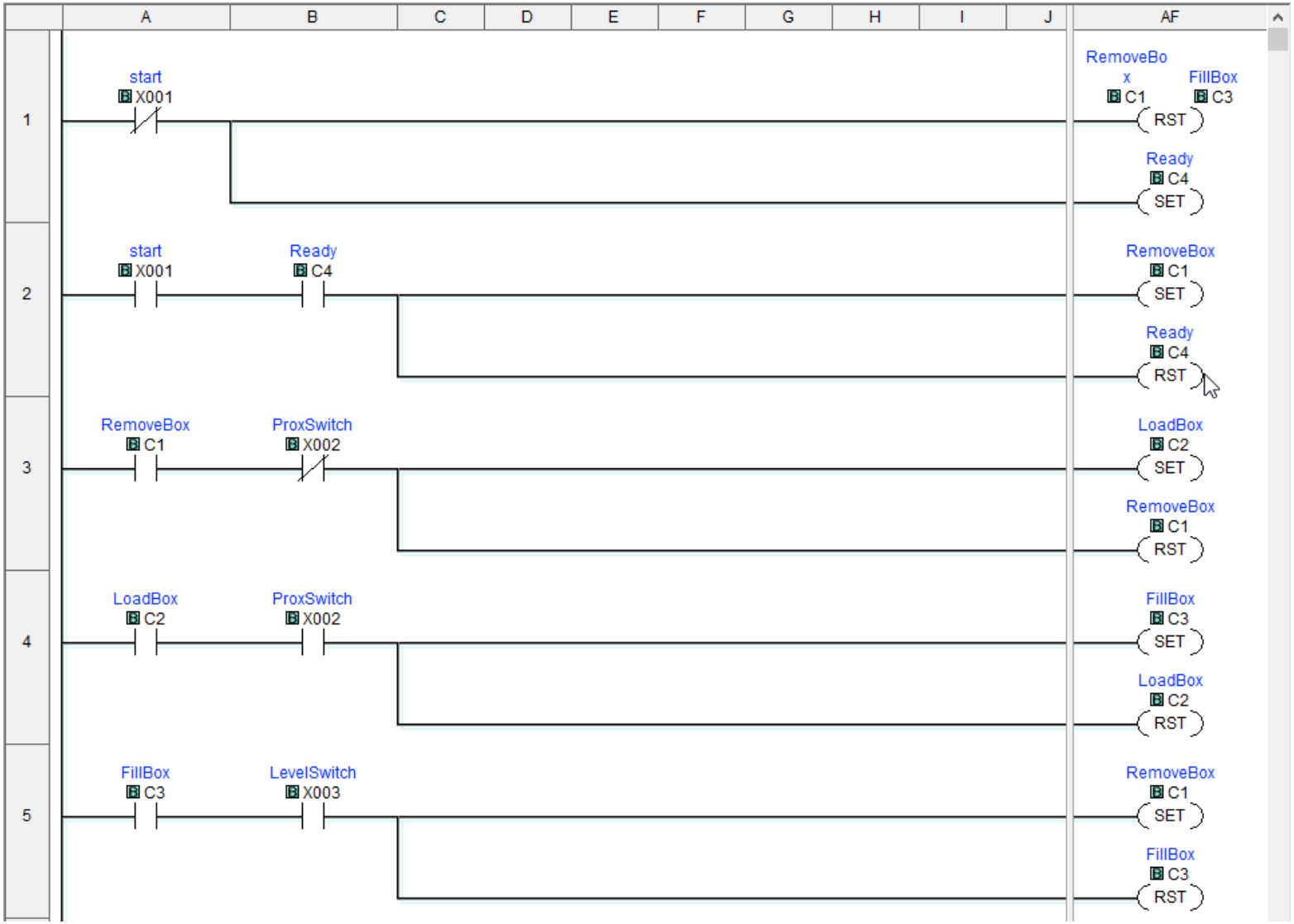
Box Filling



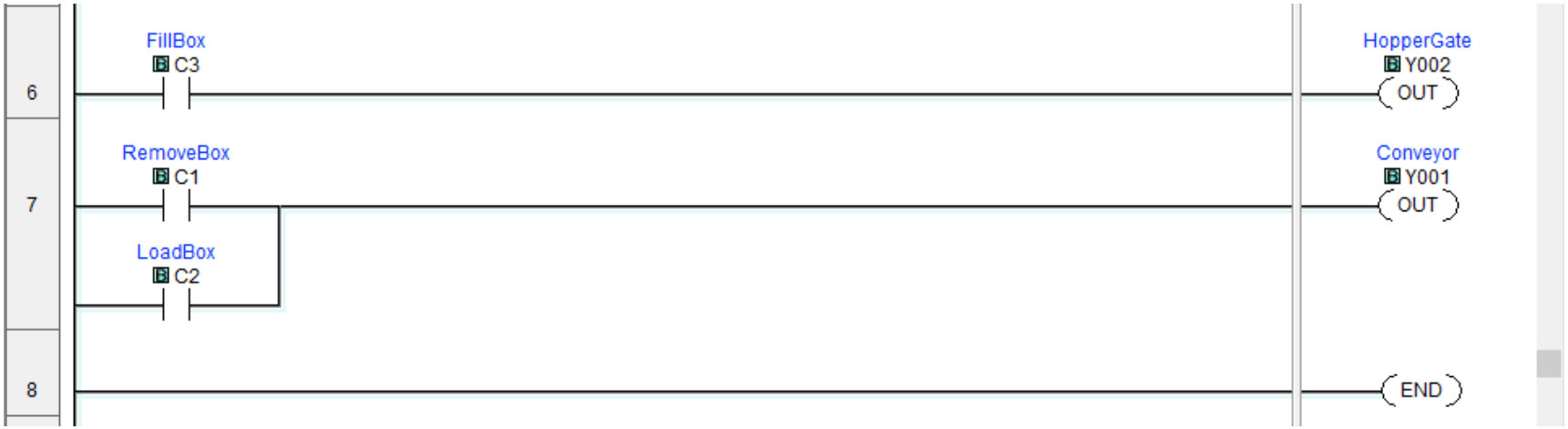
Specifications:

1) The process starts and stops with a toggle switch
2) At start, from standby, any box in the vicinity must be removed. This is the remove box state.
3) Then a new box is placed underneath of the hopper. This is the load stage. The proximity switch will announce when the new box is loaded under the hopper.
4) After a box is loaded, the process of filling the box starts. A level switch will indicate when the box is full. After the filling, the box is then removed and the entire process repeats itself.
5) There is a contact to open or close the hopper gate, and there is a contact to start and stop the conveyor.

Box Filling Solution



Box Filling Solution



Box Filling Solution: No RST/SET Complicated!

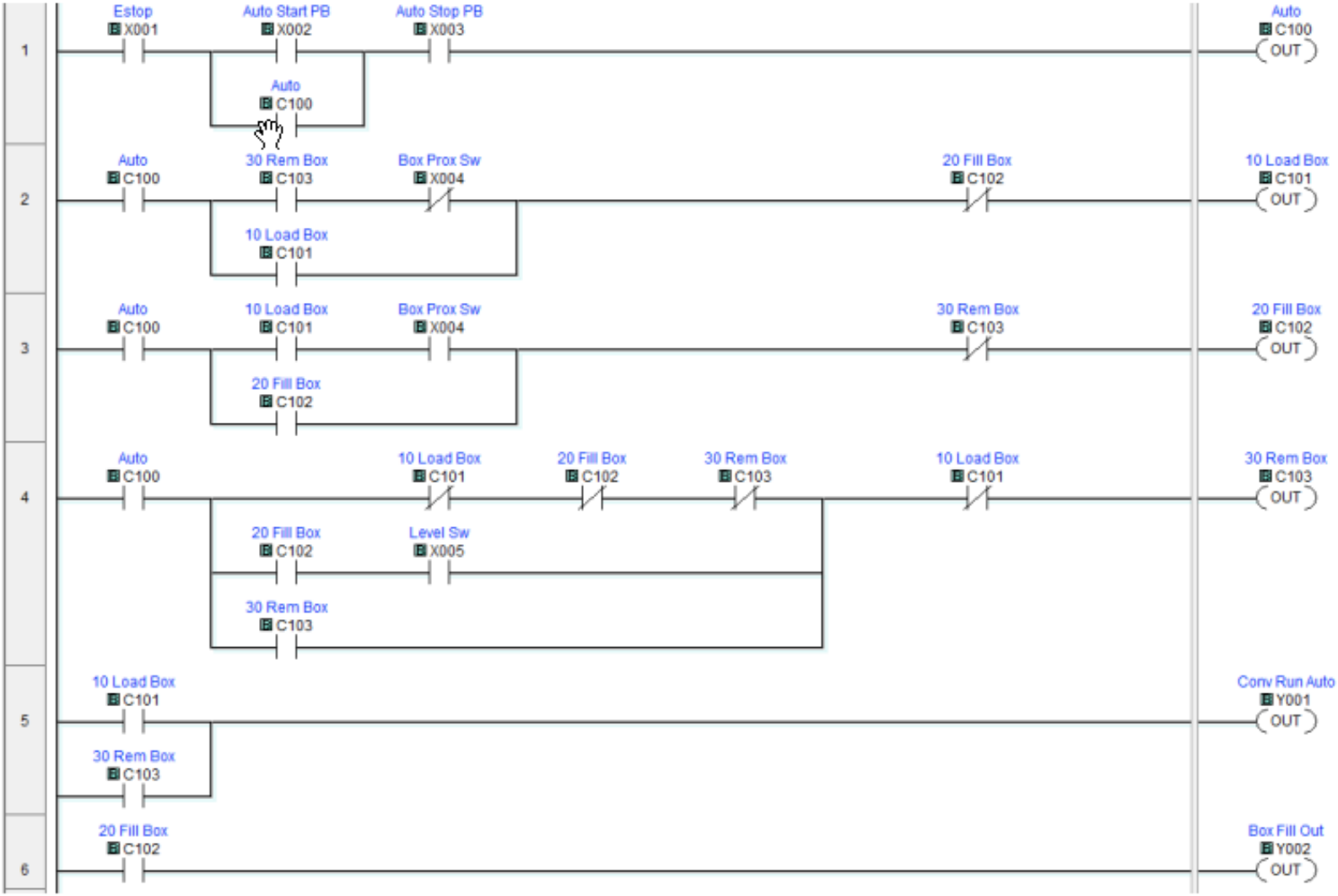


Figure 4 A three state FSM controls a box fill operation.

Discussion: The Power-On/Power-Off button

- Consider the Power-On/Power-Off push button. One push (momentary) button is used to start and to stop a machine.
- Design an FSM whose input X1 is the push button and its output is a signal Y1 that is “on” to start the machine and is “off” to shut-off the machine.

Rules to Program an FSM

- Start by designing your FSM with a bubble diagram that shows the states and transitions.
- Indicate the ENTRY, WHILE and EXIT of each state (be clear and consistent with your definitions).
- Implement your FSM in a ladder diagram. There must a be one-to-one correspondence between your FSM and your ladder diagram.
- Each state of your FSM must occupy one rung activated by a bit variable named with the same name as that of the state in the FSM. Use the C variables (Click) or the bit B3:xx/yy (Allan Bradley) variables to identify each state.
- In the top of your ladder diagram establish your initialization for power-off and power-on. The power-off (standby) and the power-on (operation) can be considered as hyper-states
- Use the SET and RESET instructions (Click) or the Latch and Unlatch instructions (Allan-Bradley) to enter and exit each state.
- Outputs may be be left for the end of the diagram as they look better there for readability and unique location.